

Polynomials

Name
Date
Course

Simplify the following polynomials, write in descending order of powers, and state the degree.

① $8x^7 - 2x^5 + 3x^7 + 6x - 9x^5 + 4$

$11x^7 - 11x^5 + 6x + 4$, $D=7$

② $-5x^3 + 3x^4 + 10 + 3x^2 + 6x - 9 + 8x^4 - 2x + 5x - 4x^2$

$11x^4 - 6x^2 + 9x + 1$, $D=4$

③ $2 - x^3 - 12x^2 + 3x^2 + 6x + 1 - 8x + x^2$

$-x^3 - 8x^2 - 2x + 3$, $D=3$

④ $8x + 3x^4 - 5x^2 + x^2 - 3x + 2x^4 + 1$

$5x^4 - 4x^2 + 5x + 1$, $D=4$

⑤ $3x^8 - 20x + 2x - 4x^5 - 2x^8 + x$

$x^8 - 4x^5 - 17x$, $D=8$

⑥ $-3x - 4 + 6x^2 + 10x^2 - 4x + 2$

$16x^2 - 7x - 2$, $D=2$

Add the following polynomials.

⑦ $(5x^3 - 2x^2 + 6x - 8) + (3x^2 + 6x - 4)$

$5x^3 - 2x^2 + 6x - 8 + 3x^2 + 6x - 4$

$5x^3 + x^2 + 12x - 12$

⑧ $(x^3 - 2x^2 + 3) + (4x^3 - 5)$

$5x^3 - 2x^2 - 2$

⑨ $(-9x^5 + 2x^3 - 6) + (4x^4 - 3x^3 + 8x^2 - 1)$

$-9x^5 + 4x^4 - x^3 + 8x^2 - 7$

⑩ $(5x + 7) + (2x^3 + 6x + 7)$

$2x^3 + 11x + 14$

$$\textcircled{11} (-x^4 + x^3 + x^2 - 3x + 7) + (-2x^3 + 4x + 8)$$

$$\boxed{-x^4 - x^3 + x^2 + x + 15}$$

$$\textcircled{12} (-6x^2 - 2x - 2) + (-4x^2 - x - 3)$$

$$\boxed{-10x^2 - 3x - 5}$$

$$\textcircled{13} (x^2 - 10) + (8x - 9) + (x^2 - 4x + 6)$$

$$\boxed{2x^2 + 4x - 13}$$

Subtract the following polynomials.

$$\textcircled{14} (4x^3 - 9x^2 - 2x + 3) - (6x^3 - 7x^2 + 2x - 10)$$

$$4x^3 - 9x^2 - 2x + 3 - 6x^3 + 7x^2 - 2x + 10$$

$$\boxed{-2x^3 - 2x^2 - 4x + 13}$$

$$\textcircled{15} (10x^4 + 3x^3 - 7) - (8x^3 + 2x^2 - 3x + 2)$$

$$10x^4 + 3x^3 - 7 + (-8x^3) - 2x^2 + 3x - 2$$

$$\boxed{10x^4 - 5x^3 - 2x^2 + 3x - 9}$$

$$\textcircled{16} (-5x^3 - 2x^2 + 6x + 5) - (3x^4 - 6x^3 + 2x^2 - 3x - 3)$$

$$-5x^3 - 2x^2 + 6x + 5 + (-3x^4 + 6x^3 - 2x^2 + 3x + 3)$$

$$\boxed{-3x^4 + x^3 - 4x^2 + 9x + 8}$$

$$\textcircled{17} (-4x^3 + x) - (-2x^3 + 6x - 9)$$

$$-4x^3 + x + 2x^3 - 6x + 9$$

$$\boxed{-2x^3 - 5x + 9}$$

$$\textcircled{18} (x^2 + 7x - 4) - (2x^4 - 8x^3 + 2)$$

$$x^2 + 7x - 4 - 2x^4 + 8x^3 - 2$$

$$\boxed{-2x^4 + 8x^3 + x^2 + 7x - 6}$$

$$\textcircled{19} (-3x^8 + 2x^5 - 7x^2 + 4) - (7x^8 - 2x^3 + x^2 - 1)$$

$$-3x^8 + 2x^5 - 7x^2 + 4 - 7x^8 + 2x^3 - x^2 + 1$$

$$\boxed{-10x^8 + 2x^5 + 2x^3 - 8x^2 + 5}$$

$$\textcircled{20} (4x^3 - 2x) - (-8x + 1)$$

$$4x^3 - 2x + 8x - 1$$

$$\boxed{4x^3 + 6x - 1}$$

$$\textcircled{21} (-2x^3 + 4x - 8) - (7x^2 + 2x + 4)$$

$$-2x^3 + 4x - 8 - 7x^2 - 2x - 4$$

$$\boxed{-2x^3 - 7x^2 + 2x - 12}$$

Multiply the following polynomials.

$$(22) 8x^2(3x^4 - 3x + 5) = 24x^6 - 24x^3 + 40x^2$$

$$(23) -2z^3(2z^4 - 8z^3 + 3z^2 - 4z - 9) = -4z^7 + 16z^6 - 6z^5 + 8z^4 + 18z^3$$

$$(24) 3y^4(y^3 - 5y^2 - y + 7) = 3y^7 - 15y^6 - 3y^5 + 21y^4$$

$$(25) -3x^8(3x^4 - 2x^3 - 7) = -9x^{12} + 6x^{11} + 21x^8$$

$$(26) -2y^8(-3y^3 + 8y^2 - 2y + 3) = 6y^{11} - 16y^{10} + 4y^9 - 6y^8$$

$$(27) -z^3(-3z^3 + 4z^2 - 10z + 2) = 3z^6 - 4z^5 + 10z^4 - 2z^3$$

$$(28) 5x(x + 2) = 5x^2 + 10x$$

$$(29) 7y(y^3 + 10y^2 + 3y - 7) = 7y^4 + 70y^3 + 21y^2 - 49y$$

$$(30) 9z^{10}(-z^4 + 8z^3 - 6) = -9z^{14} + 72z^{13} - 54z^{10}$$

$$(31) -5x^2(4x^3 - 7x + 4) = -20x^5 + 35x^3 - 20x^2$$

$$(32) -y^3(-2y^2 - 9y - 2) = 2y^5 + 9y^4 + 2y^3$$

Divide the following polynomials.

$$(33) \frac{10x^6 + 20x^5 - 16x^4 + 4x^3}{2x^2}$$

$$5x^4 + 10x^3 - 8x^2 + 2x$$

$$(34) \frac{-36y^5 - 27y^4 + 45y^3 - 81y^2}{-9y}$$

$$4y^4 + 3y^3 - 5y^2 + 9y$$

$$(35) \frac{21z^{10} + 9z^5}{3z^5}$$

$$7z^5 + 3$$

$$(36) \frac{25x^5 + 60x^4 - 35x^3 - 70x^2 + 40x}{-5x}$$

$$-5x^4 - 12x^3 + 7x^2 + 14x - 8$$

$$(37) \frac{-56x^6 + 24x^5 - 32x^4 + 40x^3}{8x^2}$$

$$-7x^4 + 3x^3 - 4x^2 + 5x$$

$$(38) \frac{18y^5 + 36y^4 - 45y^3 + 72y^2}{-9y^2}$$

$$-2y^3 - 4y^2 + 5y - 8$$

$$(39) \frac{20z^8 + 12z^6}{4z^6}$$

$$5z^2 + 3$$

$$(40) \frac{-18x^5 + 36x^4 - 60x^3 + 48x^2 - 6x}{-6x}$$

$$3x^4 - 6x^3 + 10x^2 - 8x + 1$$

$$(41) \frac{49y^5 - 28y^4 + 63y^3}{7y^3}$$

$$7y^2 - 4y + 9y^5$$

$$(42) \frac{-100z^8 + 50z^3 - 90z^2}{-10z^2}$$

$$10z^6 - 5z + 9$$